

Question:

A 3-day-old female infant presents with poor feeding, lethargy, vomiting after feeding, and seizures. Labs revealed ketoacidosis and elevated hydroxypropionic acid levels. Upon administration of parenteral glucose and protein devoid of valine, leucine, methionine, and threonine, and carnitine, the infant began to recover. Which of the following enzymes is most likely deficient in this infant?

Answer Choices:

- A. Branched-chain ketoacid dehydrogenase
- B. Phenylalanine hydroxylase
- C. Propionyl-CoA carboxylase
- D. Cystathionine synthase

Reasoning + Answer:

<think> </think><answer>A. Branched-chain keto acid dehydrogenase </answer>

